



EARLY SIGNALS PREDICT QUALITY IN IMAGE GENERATION

Bachelor's Project Thesis

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Abstract: Text-to-image diffusion models are usually evaluated only after generation is complete. This thesis investigates whether interpretable statistics from the first five denoising steps can already predict final image quality. I generated 36,600 images from 7,320 prompts using Stable Diffusion 3.5 Large, with five seeds per prompt, and scored each image with CLIP, HPSv2, and ImageReward. During generation, I recorded latent states, denoiser predictions, classifier-free guidance signals, and consistency measures, then used these prefix features for global prediction, prompt-controlled seed ranking, and early-abort simulation.

Early trajectory features contained measurable quality signal. A training-free health score performed modestly above chance for global prediction, while a supervised logistic-regression model performed better, especially for HPSv2 (ROC-AUC=0.764). However, approximately 79–84% of quality variance was prompt-specific, making within-prompt seed selection stricter than global prediction. The training-free score did not rank seeds reliably, while the supervised model ranked HPSv2 seeds with 59.1% pairwise accuracy and recovered part of the possible best-of-five improvement over random seed choice.

1 Introduction

Diffusion models are a dominant approach for text-to-image generation, producing high-quality images through an iterative denoising process (Ho et al., 2020; Rombach et al., 2022). Starting from noise, the model gradually refines a latent representation into an image aligned with a given text prompt. Although diffusion models are effective, image generation remains computationally expensive and the final quality of a generated image is not known until the denoising process is complete. Different prompts and random seeds can lead to outputs with different levels of prompt alignment and perceptual quality. As a result, some computation and user waiting time is spent on generations that ultimately receive low quality scores.

Figure 1.1 illustrates the variation that motivates this thesis: generated images can receive very different automatic quality scores, and these scores are only available after generation is complete.

Most approaches to improving generation quality either change the sampling process or select

among completed candidates. Guidance schedules, adaptive guidance, and resampling methods modify the denoising trajectory itself (Wang et al., 2024; Castillo et al., 2025; Liu et al., 2024). Candidate selection can also improve final outputs, but requires fully generating several images before choosing among them. Recent probe-based methods instead predict quality from intermediate internal features, showing that early diffusion signals can contain information about final image quality (Guo et al., 2026; Cui et al., 2026).

This thesis studies a simpler and more interpretable version of that idea: compact early-trajectory statistics, including latent movement, denoiser-prediction behaviour, classifier-free guidance behaviour, and update consistency. These signals are available before the final image exists.

This leads to the following research question:

Can early denoising trajectory signals predict the final automatic quality scores of text-to-image generations?

I test the hypothesis that the first few denois-

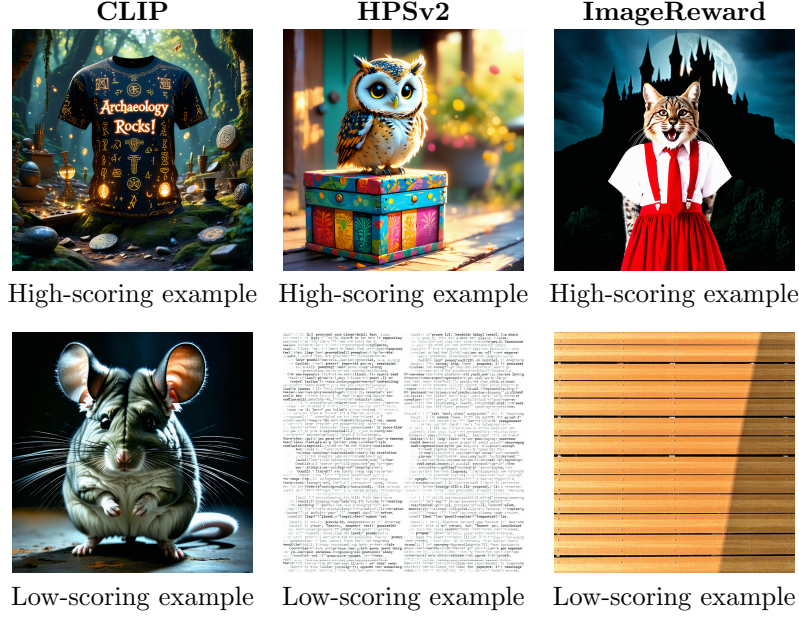


Figure 1.1: Generated examples with high and low automatic scorer values. Labels refer to scorer outputs, not human annotations, and illustrate variation in final assessed quality. The prompts are listed in Appendix A.1.

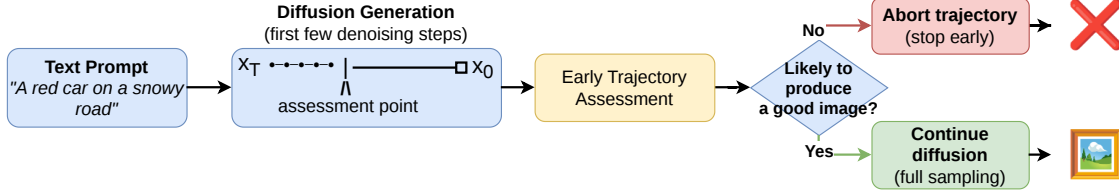


Figure 1.2: Conceptual overview of early-abort filtering. Internal trajectory signals are assessed during early denoising; predicted low-quality trajectories are stopped, while promising trajectories continue to full sampling.

ing steps contain measurable information about final image quality. More specifically, I expect compact trajectory features to predict final scorer values above chance, but I also expect this signal to be stronger for global quality filtering than for prompt-controlled seed ranking.

To test this hypothesis, I compare a hand-designed training-free health score with a supervised logistic-regression model trained on the same prefix features. The evaluation distinguishes three related tasks. First, global quality prediction asks whether early features can distinguish above-median from below-median final scorer values across all generated images. Second, prompt-

controlled seed ranking asks whether the same signals can rank different seeds for the same prompt. Third, early-abort filtering asks whether low-scoring trajectories can be identified early enough to save generation steps. This distinction is important because a score that works as a global warning signal may still fail to choose the best seed within a fixed prompt. Figure 1.2 illustrates the intended early-abort use case.

The goal of this work is therefore not to introduce a state-of-the-art early quality predictor, but to test how far compact and interpretable trajectory summaries can go. If early trajectory behaviour is predictive of final quality, such signals could help fu-

ture image-generation systems stop or deprioritise low-scoring generations before the full image is produced.

2 Literature Review

Text-to-image diffusion models generate images through an iterative denoising trajectory rather than a single forward pass. This makes it possible to inspect intermediate latent states and denoiser behaviour before the final image exists. In this thesis, these intermediate quantities are treated as potential predictors of final automatic quality scores.

2.1 Text-to-Image Diffusion and Guidance

Denoising diffusion models generate samples by gradually transforming noise into a structured output (Ho et al., 2020). Latent diffusion models perform this process in a learned latent space rather than directly in pixel space, making high-resolution generation more efficient (Rombach et al., 2022). Modern text-to-image systems use this iterative process to generate images conditioned on text prompts.

The important property for this thesis is not the full training objective of diffusion models, but the fact that generation exposes a trajectory. At each denoising step, the model has a latent state and a predicted denoising direction. These quantities can be measured before the final image is available.

Text conditioning is commonly strengthened with classifier-free guidance (Ho & Salimans, 2022). In classifier-free guidance, the model computes both a conditional prediction, based on the prompt, and an unconditional prediction. The difference between these predictions determines how strongly the prompt steers the denoising direction. For this thesis, guidance behaviour is relevant because it gives an internal signal of how the prompt affects the early trajectory.

The denoising process therefore exposes several internal signal families during inference: latent movement, denoiser predictions, guidance behaviour, and update consistency. These signals are available before the image is complete. The main question is whether they only describe the mechan-

ics of sampling, or whether they also contain information about final image quality.

2.2 Automatic Quality Evaluation

Text-to-image quality is not one-dimensional. An image can match the prompt while still being visually weak, or look plausible while missing important prompt details. For this reason, this thesis evaluates final images with three automatic scorers.

CLIP measures prompt-image alignment by embedding text and images into a shared representation space and comparing their similarity (Radford et al., 2021). In this thesis, CLIP is treated mainly as an alignment score. HPSv2 and ImageReward are used as preference-oriented quality proxies. HPSv2 is trained on human preference data for text-to-image outputs (Wu et al., 2023), while ImageReward is a learned reward model trained from human preference annotations (Xu & Liu, 2023).

These scorers are not human ground truth. They are automatic targets used to test whether early trajectory features predict the scores that finished images later receive. Throughout the thesis, “high” and “low” quality refer to high- and low-scoring images according to a given automatic scorer.

2.3 Inference-Time Improvement and Selection

Recent work frames diffusion inference as a resource-allocation problem, where additional computation can improve outputs at sampling time (Ma et al., 2025). Other approaches modify the sampling trajectory directly, for example through guidance scheduling, adaptive guidance, or resampling (Wang et al., 2024; Castillo et al., 2025; Liu et al., 2024). These methods show that the sampling trajectory can be inspected or changed during generation, but their goal is different from the present thesis.

Another common strategy is to generate several candidates and keep the best one. This can improve quality, but it requires completing several full generations. The relevant question for this thesis is whether low-scoring candidates can instead be identified early, before all sampling steps are spent.

2.4 Early Quality Prediction with Learned Probes

Recent work has directly studied early quality prediction in text-to-image diffusion models. Probe-Select predicts final image quality from intermediate denoiser activations and uses this prediction to decide which candidate generations should continue (Guo et al., 2026). Diffusion Probe uses early cross-attention statistics to forecast final image quality (Cui et al., 2026). These methods show that early internal diffusion signals can contain information about final image quality.

This thesis differs from these probe-based methods in the type of signal it studies. Rather than using rich denoiser activations or attention maps, I use compact trajectory statistics: latent movement, denoiser-prediction change, classifier-free-guidance behaviour, and update consistency. The aim is therefore not to outperform richer learned probes, but to test how much useful signal is available in simpler and more interpretable trajectory summaries.

A second difference is the evaluation framing. This thesis evaluates global quality prediction, prompt-controlled seed ranking, and early-abort filtering separately. This matters because global prediction can benefit from prompt-level difficulty, whereas seed ranking requires distinguishing between different generations of the same prompt.

2.5 Uncertainty and Reliability in Diffusion Models

Uncertainty quantification provides a related view of early quality prediction. In diffusion models, uncertainty can refer to ambiguity in the conditioning signal, variation between possible samples, or low confidence in a generated output. Previous work has studied uncertainty in diffusion models at the pixel level, sample level, and text-to-image level (Kou et al., 2024; Chan & Silva, 2024; Franchi et al., 2025).

This thesis does not estimate a full uncertainty distribution. Instead, it treats low-quality generations as unreliable outputs and asks whether simple statistics from one early denoising trajectory can predict final automatic quality scores. In this sense, the training-free health score is best interpreted as a lightweight reliability heuristic, while

the supervised prefix model tests how much additional quality-related information is present in the same trajectory features.

3 Methodology

This chapter describes the experimental pipeline used to test whether early internal denoising dynamics in a text-to-image diffusion model predict final image quality. I used Stable Diffusion 3.5 Large as a fixed image generator, recorded internal trajectory signals during inference, scored the final images with automatic quality metrics, and evaluated whether early trajectory features predicted those final scores. The methodology is organised into five parts: dataset and generation setup, prediction-task definition, trajectory feature representation, predictive models, and evaluation settings.

Figure 3.1 summarises the full pipeline, from prompt sampling and multi-seed generation to trajectory logging, final scoring, and predictive evaluation. The implementation code for generation, trajectory logging, feature extraction, scoring, modelling, and evaluation is available in the accompanying repository*.

3.1 Dataset and Generation Setup

3.1.1 Prompt Dataset

The prompt set was constructed from the Hugging Face image-preferences dataset, which contains prompts paired with text-to-image outputs and preference annotations (Data is Better Together Community, 2024). I used the original `prompt` field directly as the text input for generation.

Only prompts associated with Stable Diffusion 3.5 Large examples were retained, and duplicate prompts were removed, resulting in 7,320 unique prompts. The original preference label was not used for prompt selection, model training, or final quality labelling. In this thesis, the dataset therefore served only as a diverse prompt source. All final labels were derived from automatic scores assigned to newly generated images.

*Implementation code repository: <https://github.com/karbolak/early-signals-t2i-quality-code>

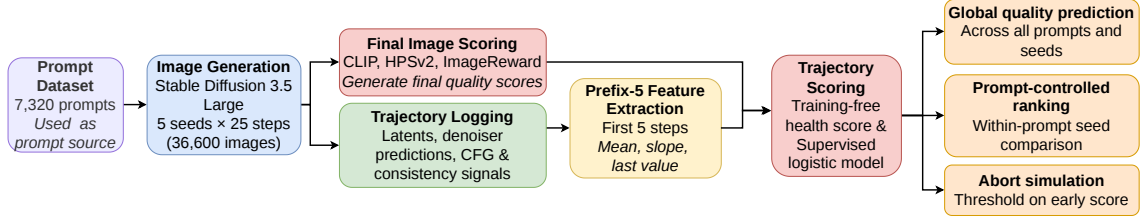


Figure 3.1: Overview of the experimental pipeline. Final images are scored after generation, while the predictors use only trajectory features recorded from the early denoising prefix.

3.1.2 Generation Model and Configuration

Stable Diffusion 3.5 Large (Stability AI, 2024; Esser et al., 2024) was used as the fixed generator for the main experiment. It was selected because its denoising process could be instrumented step by step and preliminary diagnostics showed sufficient variation across prompts and seeds. Full model-choice diagnostics are provided in Appendix B.1.

Each prompt was generated five times with different random seeds, producing 36,600 images and corresponding trajectory records. The five-seed design supported both global prediction across all generations and prompt-controlled comparisons between seeds of the same prompt. A complete generation configuration table is provided in Appendix B.2.

3.2 Prediction Task Definition

This thesis evaluates whether information available early in the denoising trajectory can predict final image quality. The prediction target was therefore defined after generation, using automatic quality scores assigned to the completed images, while the predictors used only early trajectory information.

3.2.1 Automatic Quality Scoring

Every generated image was scored with three automatic targets: CLIP, HPSv2, and ImageReward, because prompt alignment and preference-oriented quality can disagree.

CLIP was used as a prompt-image alignment score. CLIP trains an image encoder and a text encoder contrastively, so matching images and texts are close in a shared embedding space (Radford et al., 2021). In this thesis, CLIP measured how well a

generated image matched its prompt according to that embedding space. It should not be interpreted as a complete measure of perceptual or aesthetic quality.

HPSv2 was used as a human-preference-oriented score. HPSv2 is based on a CLIP-style scorer trained on the Human Preference Dataset v2, which contains pairwise human preferences over text-to-image outputs (Wu et al., 2023). Intuitively, HPSv2 keeps the prompt-image comparison structure of CLIP-like models but tunes the scalar score toward observed human preferences.

ImageReward was used as a second preference-oriented target. ImageReward is a learned reward model trained from human preference annotations for text-to-image generation (Xu & Liu, 2023). Given a prompt and image, it returns a scalar reward intended to approximate human judgement. I included it because it provides a preference signal trained differently from HPSv2. Because the metrics used different numerical scales, I compared models within each metric rather than comparing raw scores across metrics.

3.2.2 Binary Label Construction

For each quality target, I converted the continuous final scores into a binary high-scoring versus low-scoring label using a median split. Generations above the median score for a given target were labelled as high-scoring, and generations at or below the median were labelled as low-scoring. The median was used because it produced balanced classes for each target and avoided assuming that CLIP, HPSv2, and ImageReward had comparable raw score scales.

This binarisation was used for the global classification evaluation and for training the supervised

trajectory model. The resulting predicted probabilities were then used as continuous learned trajectory scores in later ranking, best-of-five, and abort-simulation analyses. Thus, the analysis did not assume that image quality is truly binary. The median split was an operational way to test whether early trajectory signals could identify generations likely to finish in the higher-scoring half for each automatic target.

3.2.3 Early-Prefix Prediction Setting

The main prediction setting used only the first five denoising steps, referred to as prefix-5. Since all generations used 25 denoising steps, prefix-5 corresponds to the first 20% of the trajectory. This prefix length was chosen because it is early enough to leave substantial computation to save in an abort setting, while still allowing several denoising updates to be observed. Longer prefixes were analysed as diagnostics in the Results and Appendix.

3.3 Trajectory Features

During generation, I recorded internal tensor-valued signals from the denoising trajectory. These signals were intermediate model states and predictions available during inference, rather than final quality scores. A denoising-step callback recorded the latent state after each step, while a wrapper around the model forward pass recorded denoiser predictions and classifier-free guidance quantities.

At denoising step t , let z_t denote the latent state, $\Delta z_t = z_t - z_{t-1}$ the latent update, and $\hat{d}_t$ the guided denoiser prediction. For classifier-free guidance, let c_t and u_t denote the conditional and unconditional denoiser predictions. The guidance difference was defined as $d_t^{\text{CFG}} = c_t - u_t$.

Table 3.1 summarises how these recorded tensors were converted into interpretable signal families. For example, root-mean-square (RMS) measured the scale of a tensor such as z_t , Δz_t , $\hat{d}_t$, or d_t^{CFG} , while cosine similarity measured whether two tensor directions agreed. Thus, x in $\text{RMS}(x)$ is not a loss term, but one of the recorded tensors or tensor differences from the denoising trajectory. Full mathematical definitions are provided in Appendix B.3.

These signal families were chosen because low-scoring generations may already differ early in the

denoising process through unstable latent movement, unusual guidance behaviour, unstable denoiser predictions, or weak agreement between predicted and actual updates.

3.3.1 Prefix Feature Extraction

The trajectory signals in Table 3.1 produced scalar time series over denoising steps. For each scalar trajectory signal, I computed three prefix summaries over the selected prefix: the mean value, the linear slope, and the final value at the end of the prefix. The slope was the coefficient of a first-degree polynomial fitted over the prefix steps. If fewer than two valid values were available, the slope was set to zero. The resulting compact prefix feature set contained 45 features, derived from 15 scalar signal types and three summaries per signal. The full feature list is provided in Appendix B.4.

3.4 Predictive Models

The extracted prefix features were used to construct two predictors. The first was a hand-designed training-free health score, intended to test whether simple interpretable rules could identify promising or problematic trajectories without fitting a model. The second was a supervised logistic-regression model, intended as an information probe for how much predictive signal was present in the same prefix features when the feature weights were learned from data.

3.4.1 Training-Free Health Score

The first predictor was a hand-designed training-free health score. It tested whether a fixed rule over early trajectory statistics could identify promising or problematic generations without learning from labelled examples.

Each feature was standardised as $z_{ik} = (x_{ik} - \mu_k) / (\sigma_k + \epsilon)$, where x_{ik} is feature k for generation i , and μ_k and σ_k are the dataset mean and standard deviation of that feature. The raw score was $\tilde{H}_i = \sum_k w_k z_{ik}$, where w_k is a fixed hand-chosen weight. The final score H_i was obtained by min-max scaling $\tilde{H}_i$ to $[0, 100]$.

The score rewarded signals hypothesised to indicate productive early denoising, such as latent volatility, denoiser-prediction change, update-

Table 3.1: Recorded trajectory signal families. Each family was converted into scalar time series and then summarised over the prefix.

Signal family and recorded quantity	Scalar summaries and interpretation
Latent magnitude Latent state z_t	RMS, standard deviation, and absolute mean measure the scale and dispersion of the latent representation.
Latent movement Latent update Δz_t	Update RMS and update-direction cosine similarity measure how much and how consistently the latent state changes across steps.
Denoiser prediction Guided prediction $\hat{d}_t$	Prediction scale, prediction-change RMS, and prediction cosine similarity measure the stability of the predicted sampling direction.
Classifier-free guidance Conditional prediction c_t , unconditional prediction u_t , and difference d_t^{CFG}	CFG divergence, relative divergence, and alignment measure how strongly and in what direction the prompt steers sampling.
Update consistency Latent update Δz_t and guided prediction $\hat{d}_t$	Update-prediction cosine similarity and consistency residual measure agreement between the predicted direction and the realised latent update.

prediction agreement, and moderate CFG divergence. It penalised signals hypothesised to indicate unstable or inconsistent denoising, such as excessive latent magnitude and high consistency residual. The exact feature groups and weight interpretation are provided in Appendix B.5.

The training-free score had two roles. First, it served as a training-free early-warning heuristic that did not require supervised training. Second, it provided a baseline against which the supervised prefix model could be compared.

3.4.2 Supervised Trajectory Model

The second predictor was a supervised logistic-regression model trained on the same prefix trajectory features as the training-free score. A separate model was trained for each quality target: CLIP, HPSv2, and ImageReward.

For a given target, the label indicated whether generation i was above the median score for that target. I denote this target-specific label as $\text{high}_i^{(q)}$, where q refers to the quality target. For each target q , the model estimated the probability $P(\text{high}_i^{(q)} = 1 \mid x_i)$, where x_i is the prefix trajectory feature vector for generation i . The model used standardised features, balanced class weights, and cross-validated predicted probabilities. These probabilities were used directly as target-specific learned trajectory scores. Thus, a higher value means that

the model assigns a higher probability that the generation will finish above the median quality score for that target.

This model served as an information probe. If the supervised model outperformed the training-free score, then the prefix trajectory features contained quality-related information that the fixed heuristic did not extract well. If both performed similarly, then the hand-designed score captured much of the useful signal.

3.5 Evaluation Settings

The predictors were evaluated under four complementary settings. Global prediction tested whether early trajectory scores separated high- and low-scoring generations across the full dataset. Prompt-controlled evaluation removed prompt-level difficulty and tested whether scores ranked seeds within the same prompt. Best-of-five selection was treated separately as a practical seed-selection summary. Abort simulation tested whether early scores could be used to stop likely low-scoring generations before completing the full denoising trajectory.

3.5.1 Global Predictive Evaluation

The global evaluation tested whether early trajectory scores predicted above-median final quality across all 36,600 generations. For each target, I reported ROC-AUC and PR-AUC for the training-

free score and supervised model. This analysis measured global predictive signal, but did not separate prompt difficulty from seed-specific quality.

3.5.2 Prompt-Controlled Evaluation

The prompt-controlled evaluation tested whether early trajectory scores captured seed-level differences rather than only prompt-level difficulty. Since each prompt was generated five times, the five generations for the same prompt could be compared against each other.

The prompt-controlled evaluation had three components. First, variance decomposition estimated how much quality-score variance came from differences between prompts and how much remained between seeds of the same prompt. Second, prompt-centred correlations measured whether deviations from a prompt’s average early score were associated with deviations from that prompt’s average final quality. This removed prompt-level difficulty and tested seed-level association directly. Third, within-prompt pairwise ranking accuracy compared all pairs of seeds for each prompt. A pair was concordant when the generation with the higher predicted score also had the higher true quality score, giving pairwise accuracy $N_{\text{concordant}} / (N_{\text{concordant}} + N_{\text{discordant}})$.

For supervised prompt-controlled prediction, I used grouped cross-validation by prompt. All five generations belonging to a prompt were assigned to the same fold, preventing prompt-level leakage between train and test splits.

Prompt-controlled evaluation was needed because global prediction can be driven by prompt difficulty. A score may separate generally easy prompts from generally hard prompts without identifying which seed is better for a fixed prompt. I therefore evaluated seed-level behaviour using prompt-centred correlations and within-prompt pairwise ranking, and summarised the practical seed-selection implication with a best-of-five analysis.

3.5.3 Best-of-Five Selection

The best-of-five analysis translated prediction performance into a seed-selection task. For each prompt p , let G_p be the set of five generated images, and let q_i be the final quality score of generation i .

The predicted score was used to select one generation from the five available seeds.

The selected generation was compared with two prompt-specific baselines. The random baseline was the average true quality of the five generations for the prompt, $Q_{\text{random}}(p) = |G_p|^{-1} \sum_{i \in G_p} q_i$. This represents the expected quality of choosing one seed at random. The oracle baseline was the best true quality available among the five generations, $Q_{\text{oracle}}(p) = \max_{i \in G_p} q_i$. This represents the best possible selection if the final quality scores were already known.

The lift over random was the average difference between selected quality and random expected quality. The fraction of oracle gain divided this lift by the average gap between the oracle and random baselines. This measured how much of the theoretically available best-of-five improvement was recovered by the selection rule.

3.5.4 Abort Simulation

The final evaluation considered the early-abort use case. Each generation was scored after prefix-5. For an early score s_i and threshold τ , generations with $s_i < \tau$ were treated as aborted and generations with $s_i \geq \tau$ were retained. Retained quality was the mean final quality of the retained generations, and quality lift was this retained mean minus the mean quality of all generations. Formal definitions are provided in Appendix C.6.

The simulation was performed offline: all images were fully generated first, and the abort rule was applied retrospectively. Since the decision was made after five of 25 sampling steps, each aborted generation was counted as saving 80% of its sampling steps. The learned-score threshold was matched to the selected training-free abort rate.

4 Results

Following the feature-extraction procedure described in Section 3, I evaluated whether early denoising trajectories predicted final automatic quality scores. The results are organised around five questions: whether the quality metrics behaved consistently enough to support separate target-specific analyses, whether quality variation was mainly prompt-level or seed-level, whether pre-

Table 4.1: Pairwise Pearson correlations between automatic quality metrics. Values above the diagonal are raw correlations; values below the diagonal are prompt-centred correlations, computed after removing each prompt’s mean score.

	CLIP	HPSv2	ImageReward
CLIP	–	0.370	0.383
HPSv2	0.145	–	0.557
ImageReward	0.232	0.470	–

fix features predicted global quality, whether they supported prompt-controlled seed selection, and whether they could be used for early-abort filtering.

4.1 Quality-Metric Sanity Checks

Before evaluating early trajectory predictors, I first checked how the automatic quality metrics related to each other and how much quality variation was available within prompts. These checks were important because the thesis compares three target-specific prediction problems rather than treating image quality as a single scalar quantity.

4.1.1 Metric agreement

The three automatic quality metrics measured related but partially different aspects of the generated images. CLIP was treated as a prompt–image alignment score, while HPSv2 and ImageReward were treated as preference-oriented scores. For this reason, all predictive analyses were performed separately for each target rather than by combining raw scores across metrics. Full distribution summaries are provided in Appendix C.1.

Table 4.1 shows that HPSv2 and ImageReward were the most aligned pair, both in raw scores ($r=0.557$) and after prompt centring ($r=0.470$). CLIP was more distinct, especially after prompt centring. Since prompt centring removes each prompt’s mean score, the lower prompt-centred correlations show that metric agreement was weaker when comparing seeds within the same prompt. Additional within-prompt metric-agreement diagnostics are reported in Appendix C.2.

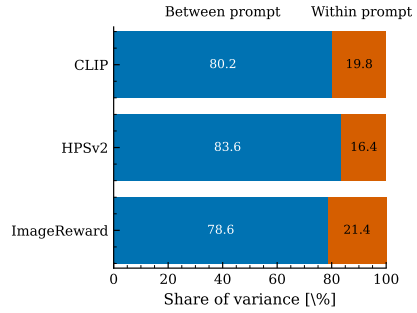


Figure 4.1: Prompt-level variance decomposition. Most quality-score variance came from prompt identity, with a smaller but non-trivial share remaining between seeds of the same prompt.

4.1.2 Prompt-level and seed-level variation

A second prerequisite was to determine whether quality variation was mostly due to prompt identity or to random seed variation within the same prompt. Figure 4.1 decomposes quality-score variance into between-prompt and within-prompt components. Across all three metrics, most variation came from prompt identity: between-prompt variance accounted for 80.2% of CLIP variance, 83.6% of HPSv2 variance, and 78.6% of ImageReward variance.

The remaining within-prompt component, between 16.4% and 21.4%, showed that seed selection could improve quality in principle, but that it was a stricter task than global prediction. A model can achieve useful global prediction by learning signals associated with prompt difficulty, even if it cannot distinguish seeds of the same prompt. For this reason, the following sections evaluate global prediction, prompt-controlled seed ranking, best-of-five selection, and early-abort filtering as related but separate settings. Additional prompt-controlled diagnostics and per-prompt breakdowns are reported in Appendix C.5.

4.2 Early-Trajectory Prediction

I first evaluated global median-split quality prediction across all 36,600 generations. This task asked whether early trajectory scores could distinguish above-median from below-median final quality for each automatic scorer. Figure 4.2 compares the

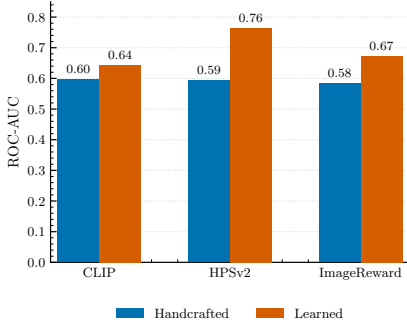


Figure 4.2: Global median-split quality prediction from prefix-5 trajectory features. The learned model outperformed the training-free health score, especially for HPSv2.

hand-designed training-free score with the supervised logistic-regression model trained on the same prefix-5 features.

The training-free score performed above chance for all three targets, with ROC-AUC values around 0.58–0.60. The supervised model extracted more signal from the same feature set, reaching ROC-AUC values of 0.643 for CLIP, 0.764 for HPSv2, and 0.672 for ImageReward. This comparison shows that compact prefix-5 trajectory features contain non-trivial information about final automatic quality after only five sampling steps, with the clearest target-specific result for HPSv2.

4.2.1 Effect of prefix length

I also evaluated how prediction changed as longer prefixes were used. Figure 4.3 shows that the first five denoising steps already captured much of the supervised predictive signal. For the supervised model, prefix-5 achieved 97.0% of the peak ROC-AUC for CLIP, 95.4% for HPSv2, and 97.5% for ImageReward. Later steps improved performance only moderately relative to the extra computation required.

The training-free score behaved differently: it peaked around five steps and declined when later summaries were included. Together, these results support prefix-5 as a useful operating point, early enough to leave substantial computation to save while still preserving most of the supervised predictive signal.

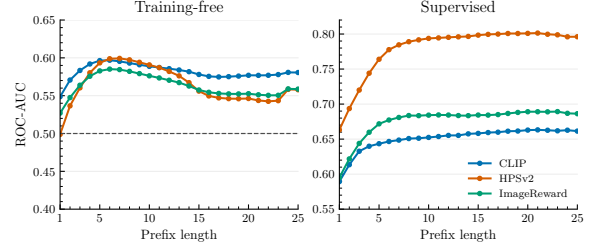


Figure 4.3: ROC-AUC by prefix length. Prefix-5 retained most supervised predictive signal, while the training-free score peaked early.

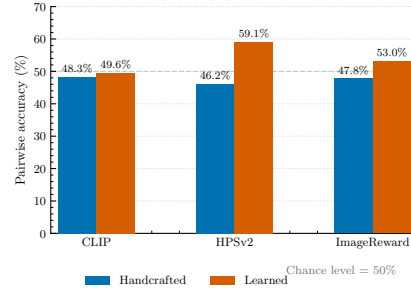


Figure 4.4: Within-prompt pairwise ranking accuracy. The dashed line marks chance performance; only the learned HPSv2 score clearly ranked seeds above chance.

4.3 Prompt-Controlled Selection

I next evaluated whether early trajectory scores could distinguish between different seeds of the same prompt. This is a stricter setting than global prediction, because prompt-level differences are held fixed and the score must rank alternative generations for a given prompt.

4.3.1 Within-prompt seed ranking

For the pairwise-ranking test, all pairs of seeds within each prompt were compared. A method was correct when the seed with the higher early trajectory score also had the higher final quality score. Figure 4.4 shows the resulting within-prompt pairwise accuracy. Additional prompt-controlled diagnostics are provided in Appendix C.5.

The training-free score did not work reliably as a seed-ranking score. It performed below chance for all three metrics: 48.3% for CLIP, 46.2% for HPSv2, and 47.8% for ImageReward. The super-

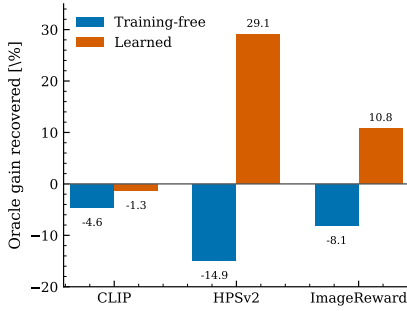


Figure 4.5: Oracle best-of-five improvement recovered by each early trajectory score. 0% corresponds to random seed selection and 100% to oracle seed selection.

vised score was also near chance for CLIP, but reached 53.0% for ImageReward and 59.1% for HPSv2. This suggests that the seed-level signal was not uniform across quality metrics. In particular, early trajectory features did not meaningfully help with prompt-image alignment as measured by CLIP, but they did support ranking for preference-oriented quality, especially HPSv2.

4.3.2 Best-of-five selection

Best-of-five selection evaluated the same question in a practical seed-selection setting. For each prompt, the early trajectory score was used to select one of the five generated images. The selected image was compared with two prompt-specific baselines: random seed selection and oracle seed selection. The random baseline represents the expected quality of choosing one of the five seeds at random. The oracle baseline uses the final quality scores to choose the best of the five generations, and therefore represents the maximum possible improvement available from seed selection.

The reported oracle-gain fraction measures how much of the available improvement over random seed selection was recovered by the early trajectory score. A value of 0 means that the score performed like random seed selection, a value of 1 means that it matched the oracle selector, and negative values mean that the selected seed was worse than random on average. Figure 4.5 reports this quantity for each score.

The supervised HPSv2 trajectory score recovered 29.1% of the available oracle improvement,

while the supervised ImageReward score recovered 10.8%. Selection based on CLIP did not improve over random seed selection. The training-free score was negative for all three targets, confirming that it should not be interpreted as a prompt-controlled seed selector.

Together, the pairwise-ranking and best-of-five results suggest that seed selection mainly affected preference-oriented image quality rather than prompt-image alignment. CLIP scores were difficult to improve by selecting among seeds of the same prompt, whereas HPSv2, and to a lesser extent ImageReward, showed positive seed-selection signal. The useful seed-level variation therefore appeared primarily in subjective or preference-oriented quality, not in alignment.

4.4 Early-Abort Filtering

The weak seed-ranking performance of the training-free score did not rule out early-abort filtering. In an abort setting, the score does not need to choose the best seed for each prompt. It only needs to identify generations that are likely to fall in the lower-scoring part of the full dataset.

I compared abort-style filtering at a matched abort rate. For the training-free score, I used the threshold $H < 25$: generations with health score below 25 were treated as aborted, and generations with $H \geq 25$ were retained. This aborted 31.0% of generations. Because the decision was made after five of 25 denoising steps, each aborted generation would save approximately 80% of its sampling steps, corresponding to an estimated total saving of 24.8%. For the supervised score, I chose the threshold on the predicted probability so that it aborted the same fraction of generations. Thus, the comparison matched the abort rate, but each method decided which generations to abort using its own early score.

Figure 4.6 reports retained-quality lift. This is the mean final quality of the retained set minus the mean final quality over all generations. A positive lift therefore means that the early-abort rule removed lower-scoring generations often enough to increase the average final quality of the generations that remained. At the matched abort rate, the training-free score improved retained quality by 1.3% for CLIP, 1.3% for HPSv2, and 4.8% for ImageReward. The supervised score produced larger

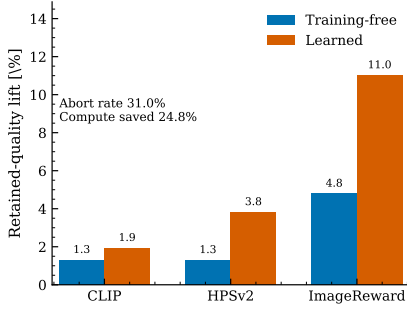


Figure 4.6: Matched-abort-rate early-abort simulation. Both scores aborted approximately 31.0% of generations; quality lift is the retained-set mean minus the full-dataset mean.

lifts: 1.9% for CLIP, 3.8% for HPSv2, and 11.0% for ImageReward.

The threshold sweep for the training-free score showed the expected trade-off. Increasing the threshold aborted more generations and increased the mean quality of the retained set, but also left fewer completed outputs. At threshold 30, the abort rate increased to 45.9%, with retained-quality lifts of 1.7% for CLIP, 1.8% for HPSv2, and 6.5% for ImageReward. Full threshold values are reported in Appendix C.6.

These gains were consistent with the prompt-controlled results. Early-abort filtering is a global filtering problem, so it can benefit from prompt-level quality differences. A score can therefore be useful for aborting likely low-scoring generations even if it cannot reliably choose the best seed within a fixed prompt. In this sense, early-abort filtering is a weaker but more practical requirement than exact within-prompt seed ranking.

4.5 Feature Diagnostics

Table 4.2 summarises representative feature correlations. The table separates global associations from prompt-controlled associations. Prompt-controlled correlations were computed after removing prompt-level means, so they describe seed-level relationships within the same prompt rather than differences between easier and harder prompts.

The main insight from Table 4.2 is that global prediction and prompt-controlled seed ranking were associated with different trajectory patterns.

At the raw dataset level, HPSv2 was most strongly associated with classifier-free guidance features, especially CFG relative divergence. This suggests that global quality prediction partly depended on how strongly the prompt-conditioned trajectory separated from the unconditional trajectory.

After controlling for prompt effects, the pattern changed. CLIP showed no clear prompt-controlled feature pattern: its strongest feature correlations were very weak and mixed in sign, with the largest absolute correlation equal to 0.052. This was consistent with the near-chance CLIP seed-ranking result. HPSv2 showed stronger prompt-controlled associations, especially with early latent stability: seeds with lower early latent volatility tended to receive higher HPSv2 scores within the same prompt. ImageReward followed the same direction but more weakly.

These patterns help explain why the supervised model outperformed the hand-designed score: it could learn target-specific feature weights, while the training-free score used one fixed weighting scheme. Full coefficient and feature-correlation diagnostics are provided in Appendix C.7.

5 Discussion

The main contribution of this thesis is to show that compact features from the first few diffusion sampling steps contain information about final automatic image quality. This information was not strong enough to solve all early selection problems, but it was useful for early quality estimation, especially in the global filtering setting. The supervised prefix model provided the clearest evidence that the early trajectory features were informative, while the training-free health score showed how far a simple hand-designed heuristic could go without labelled training examples.

This distinction is important for interpreting the results. The thesis does not show that a fixed training-free score can fully replace supervised quality prediction. Instead, it shows that early trajectory statistics are a meaningful object of study: they contain signal about final quality, they can be used before a generation is complete, and they support early-abort filtering more naturally than prompt-controlled seed selection.

Table 4.2: Representative feature-diagnostic correlations. Prompt-controlled values remove prompt-level means and therefore describe within-prompt seed-level associations.

Setting	Main feature pattern	Correlation
Global HPSv2	CFG relative divergence	$\rho \approx 0.523$
Prompt-controlled HPSv2	Early latent volatility	$\rho \approx -0.30$ to -0.33
Prompt-controlled ImageReward	Early latent volatility	$\rho \approx -0.13$
Prompt-controlled CLIP	Weak, mixed-sign feature correlations	$\max \rho = 0.052$

5.1 Early Trajectories Contain Quality-Related Signal

The supervised prefix model predicted final quality above chance for CLIP, HPSv2, and ImageReward using only features from the early sampling trajectory. This supports the hypothesis that the first few generation steps already contain measurable information about the final automatic quality score. The prefix-length analysis strengthens this conclusion: prefix-5 recovered most of the predictive signal available to the supervised model, while still leaving most of the sampling process unused.

This result complements richer probe-based approaches such as Probe-Select (Guo et al., 2026) and Diffusion Probe (Cui et al., 2026). Probe-Select reports early evaluation at 20% of the trajectory and more than 60% sampling-cost reduction, while Diffusion Probe reports PCC above 0.7 and ROC-AUC above 0.9 across several settings. These numbers are not directly comparable to the present results because the methods use different representations, targets, models, and evaluation protocols. Instead, they position this thesis as a simpler point in the design space: compact, interpretable trajectory summaries are weaker than richer learned probes, but still contain useful early quality signal.

The practical implication is that quality estimation does not necessarily need to wait until an image is fully generated. A model or heuristic can inspect the early trajectory and make a preliminary judgement about whether a generation is likely to finish with a low automatic quality score. This is the strongest use case supported by the experiments.

5.2 Global Filtering and Prompt-Controlled Ranking

The experiments confirm why the distinction between global filtering and prompt-controlled seed ranking matters. Global prediction benefited from prompt-level differences: some prompts were generally easier or harder for the generator, and this made their outputs easier to classify across the full dataset. Once prompt identity was controlled, the task became much harder.

This does not make the global result uninteresting. For early-abort filtering, the system does not need to select the best seed for a prompt. It needs to identify generations that are likely to finish with low quality, so that generation steps can be focused on more promising trajectories. In that setting, prompt-level difficulty is part of the useful signal rather than a confound to remove.

5.3 Why a Training-Free Score Matters

The training-free health score is desirable because it does not require labelled generated examples, re-training for each quality metric, or a fitted model. It can be computed directly from the trajectory using fixed feature weights. This makes it interpretable, easy to reproduce, and closer to a lightweight scoring rule that could be inserted into a generation pipeline without first collecting a large labelled dataset.

The experiments show both the value and the limitation of this approach. The training-free score achieved modest global prediction performance and improved retained quality in the early-abort simulation, while the supervised model extracted substantially more signal from the same features. The gap therefore reflects the fixed weighting scheme, not an absence of trajectory signal.

5.4 Metric-Dependent Quality Prediction

The metric-dependent results show that early trajectory features do not predict every notion of image quality equally well. HPSv2 was the clearest target, while CLIP was weaker, especially in prompt-controlled analyses. One interpretation is that latent movement, guidance behaviour, and update consistency are more related to broad preference-oriented quality than to whether specific prompt semantics appear in the final image. This limits the claim: the thesis does not show that early trajectory features predict all definitions of image quality equally well, but that they are informative for several automatic targets, with the clearest evidence for HPSv2.

5.5 Prompt-Level and Trajectory-Level Signals

The supplementary comparison with the prompt-uncertainty model of Yang (2026), reported in Appendix D, helps place the trajectory-based approach in context. Prompt-only uncertainty is available before generation starts, while trajectory-level information becomes available only after sampling begins. In this dataset, prompt uncertainty had only weak associations with final quality: at the prompt level, correlations with mean final scores were close to zero for CLIP ($\rho = -0.050$), HPSv2 ($\rho = -0.053$), and ImageReward ($\rho = 0.017$). This suggests that prompt-level uncertainty and trajectory-level behaviour capture different information, and supports treating the trajectory signal as complementary rather than redundant. The trade-off is interpretability and simplicity rather than maximum predictive accuracy.

5.6 Implications for Early-Abort Filtering

The clearest practical implication is early-abort filtering. After a short prefix, a generation can be scored and low-scoring trajectories can be stopped before all sampling steps are spent. The offline simulation suggests that this can raise the mean quality of retained images while saving a substantial fraction of generation steps.

This use case fits the empirical results better than best-of-five seed selection. Early-abort filtering only requires a score that separates generally promising from generally unpromising generations. It does not require the score to perfectly rank small differences between seeds for the same prompt. The training-free score is therefore most naturally interpreted as a global filtering heuristic, while the supervised model provides evidence about the amount of additional signal available in the same features. This distinction also helps explain why the training-free score can be useful in the abort setting despite failing as a within-prompt seed-ranking score.

The current evaluation remains an offline simulation: all images were generated first, and the abort rule was applied afterwards. It therefore estimates the quality-compute trade-off under a prefix-based stopping rule, but does not establish the behaviour of a complete interactive generation system.

5.7 Limitations

The main limitation is that final quality was measured with automatic metrics rather than human judgements. CLIP, HPSv2, and ImageReward enable large-scale evaluation, but they are still proxies, so the median-split labels should be interpreted as relative labels within this generated dataset rather than absolute definitions of good and bad images. The study also used one generator, one sampling configuration, and one prompt distribution, so the relationship between trajectory features and quality may change for other models, samplers, guidance settings, or prompt sets. Finally, the two predictors involve a trade-off: the training-free score is interpretable and label-free but fixed, while the supervised model is stronger but requires labelled generated examples for each target. This limits the claim to automatic-score prediction rather than direct human preference prediction.

5.8 Future Work

The most important next step is to test whether the same trajectory features predict human judgements. This would clarify whether the HPSv2 and ImageReward results correspond to perceived image quality and whether early trajectory scoring

remains useful when quality is measured directly by users.

A second direction is cross-model validation. Repeating the analysis across different text-to-image generators, samplers, guidance settings, and prompt distributions would test whether the observed relationships are specific to Stable Diffusion 3.5 Large or reflect a more general property of diffusion sampling trajectories.

A third direction is to improve the training-free score itself. The current score uses one fixed weighting scheme, but the results suggest that global filtering and seed-level ranking depend on different trajectory properties. A more effective training-free method could be designed specifically for early-abort filtering, using the current results as evidence about which feature families are most informative.

Overall, this thesis shows that early trajectory analysis is a promising way to estimate image quality before generation is complete. The strongest evidence is for global early-abort filtering: compact features from the first few sampling steps can identify lower-scoring generations early, making quality-aware and more efficient diffusion inference possible.

6 Conclusion

In this thesis, I investigated whether early denoising trajectories in text-to-image diffusion contain information about final image quality. I used Stable Diffusion 3.5 Large as a fixed generator, generated 36,600 images from 7,320 prompts, and recorded internal trajectory signals during inference. The analysis focused on whether features from the first five denoising steps could predict final CLIP, HPSv2, and ImageReward scores.

The results show that early trajectory features contain measurable quality-related information. The supervised prefix model predicted final quality above chance for all three targets, with the clearest result for HPSv2. The first five sampling steps already contained most of the predictive signal available to the supervised model, making prefix-5 a plausible point for early quality estimation.

The results also show that this signal is easier to use for global filtering than for exact prompt-controlled seed selection. Most quality vari-

ation came from differences between prompts, rather than differences between seeds for the same prompt. Early trajectory scores are therefore most naturally suited to identifying generally promising or unpromising generations, rather than solving all seed-level ranking decisions within a fixed prompt.

This thesis contributes a trajectory-level framework for studying early quality prediction in text-to-image diffusion. It defines compact trajectory features based on latent movement, denoiser behaviour, classifier-free guidance, and update consistency. It also compares a hand-designed training-free health score with a supervised prefix model trained on the same early trajectory features. This comparison separates two questions: whether early trajectories contain quality-related information, and whether a simple heuristic can extract that information effectively.

As a practical contribution, I also release an open-source prototype package[†] for early-abort filtering with Stable Diffusion 3.5 Large. The package wraps the generator, records early trajectory signals, and supports prefix-based abort decisions.

The method is therefore best understood as an early diagnostic signal rather than a replacement for final quality evaluation: it can help decide which trajectories are worth continuing, but the strongest evidence is for global filtering rather than exact seed-level selection.

Overall, the results support early trajectory analysis as a useful approach for quality estimation in diffusion-based image generation. They show that information about final automatic quality is already present early in the sampling process, and that this information can support global filtering decisions before a full generation is complete. This makes early trajectory scoring a promising direction for more efficient and quality-aware diffusion inference.

[†]DiffGate: <https://github.com/karbolak/DiffGate>

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A Prompts Used for the Figure 1.1 Examples

Table A.1 lists the original prompts used to generate the examples shown in Figure 1.1. The prompts are reproduced as they appeared in the prompt source and were not rewritten or cleaned manually.

Table A.1: Prompts used for the high- and low-scoring examples in Figure 1.1.

Metric	Example	Prompt ID	Seed	Prompt
CLIP	High	136	147349	a magical t-shirt emblazoned with ‘Archaeology Rocks!’ in glowing runes, surrounded by floating ancient artifacts and mystical symbols in a fantastical forest, bathed in soft golden hour light, with rich textures and vibrant color palette, high resolution, sharp focus
CLIP	Low	6002	6013347	a chinchilla as dhalsim from street fighter, captured in vivid, high-contrast Ektar film, 4k, ultra-realistic, detailed focus, with dramatic chiaroscuro lighting, dynamic perspective, and rich textures, by artgerm and greg rutkowski
HPSv2	High	1645	1656348	a vibrant manga illustration of a cute owl perched on a colorful box, with soft golden hour light, rich textures, and a serene mood, high resolution, photorealistic, rule of thirds composition
HPSv2	Low	3970	3981346	It seems like the prompt you provided contains a lot of repetitive and nonsensical text. Could you please clarify what specific information or task you need help with? I’m here to assist you, and a clear and concise prompt will help me provide the best possible response.
ImageReward	High	2281	2292347	Bobcat in red dress, white shirt with red exclamation, dark castle background, vibrant colors, dynamic composition.
ImageReward	Low	5791	5802347	A wooden deck in sharp, vibrant detail with natural light.

B Supplementary Methods

B.1 Model-Choice Diagnostics

Stable Diffusion 3.5 Large was selected before the final trajectory analysis because it provided an instrumentable denoising trajectory, supported consistent large-scale generation, and produced sufficient variation in automatic quality scores across prompts and seeds. The diagnostics in Figure B.1 were used only to justify the generator choice; they are not intended as a benchmark of Stable Diffusion 3.5 Large against FLUX, another contemporary text-to-image model family.

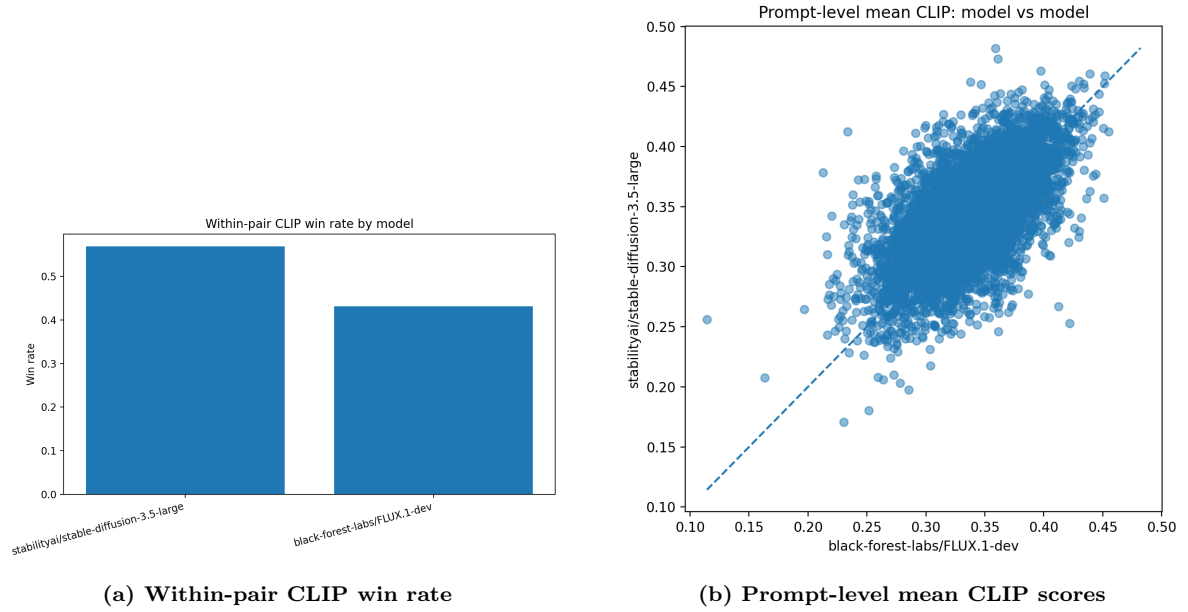


Figure B.1: Preliminary model-choice diagnostics comparing Stable Diffusion 3.5 Large with FLUX. The within-pair win-rate plot showed that Stable Diffusion 3.5 Large more often achieved the higher CLIP score for matched prompts, while the prompt-level scatter showed that both models followed a broadly similar prompt-difficulty structure. These diagnostics were used only to select a stable generator for the trajectory experiment, not as a final model benchmark.

B.2 Generation Configuration

Table B.1: Generation configuration for the main experiment.

Parameter	Value
Generator	Stable Diffusion 3.5 Large
Prompts	7,320
Seeds per prompt	5
Generated images	36,600
Resolution	1024×1024
Denoising steps	25
Guidance scale	7.0
Base seed	12345
Seed stride	1000

B.3 Trajectory Signal Definitions

This appendix gives the full definitions of the recorded trajectory signals. Let z_t denote the latent state at denoising step t , $\hat{d}_t$ denote the guided denoiser prediction, u_t denote the unconditional prediction, c_t denote the conditional prediction, and s denote the classifier-free guidance scale. In the main experiment, $s = 7.0$. The tensor x in these definitions refers to one of the recorded denoising quantities, such as a latent state, latent update, denoiser output, or guidance difference.

For a tensor x with N elements, the root-mean-square

$$\text{RMS}(x) = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}.$$

I also recorded standard deviation and mean absolute value for selected tensor-valued signals.

The latent update was

$$\Delta z_t = z_t - z_{t-1},$$

and latent volatility was defined as

$$\text{Volatility}_t = \text{RMS}(\Delta z_t).$$

Cosine similarity between two tensors a and b was

$$\cos(a, b) = \frac{a^\top b}{\|a\| \|b\| + \epsilon},$$

where ϵ is a small numerical constant. I used cosine similarity to measure agreement between successive latent updates, successive denoiser predictions, conditional and unconditional predictions, and latent updates and denoiser predictions.

Classifier-free guidance used the difference

$$d_t^{\text{CFG}} = c_t - u_t,$$

and the guided prediction

$$g_t = u_t + s d_t^{\text{CFG}}.$$

The recorded guidance signals were

$$\text{CFGDivRMS}_t = \text{RMS}(d_t^{\text{CFG}}),$$

$$\text{CFGDivRel}_t = \frac{\text{RMS}(d_t^{\text{CFG}})}{\text{RMS}(g_t) + \epsilon},$$

$$\text{CFGAlign}_t = \cos(u_t, c_t),$$

and

$$\text{GuidedMinusCondRMS}_t = \text{RMS}(g_t - c_t).$$

The denoising-consistency residual measured how well the prediction direction explained the observed latent update. The best scalar projection was

$$\alpha_t = \frac{\Delta z_t^\top \hat{d}_t}{\hat{d}_t^\top \hat{d}_t + \epsilon}.$$

The normalised residual was

$$r_t = \frac{\|\Delta z_t - \alpha_t \hat{d}_t\|}{\|\Delta z_t\| + \epsilon}.$$

Lower values of r_t indicate that the denoiser prediction direction better explained the actual update direction.

B.4 Prefix Feature Families

The prefix-5 feature set summarised the early trajectory using mean, slope, and final-prefix value for each selected signal. The main compact-prefix analysis used 15 scalar signal types, giving 45 features. Some supplementary analyses retained additional CFG-derived summaries, but the main compact-prefix analysis used the 45-feature set described here.

Table B.2: Feature families summarised over prefix-5. Each scalar signal was summarised by mean, slope, and last value.

Family	Example signals	Prefix summaries
Latent magnitude	RMS, standard deviation, absolute mean	mean, slope, last
Latent update	volatility, update cosine	mean, slope, last
Denoiser prediction	RMS, standard deviation, absolute mean	mean, slope, last
Denoiser change	delta RMS, previous-step cosine	mean, slope, last
CFG behaviour	divergence, relative divergence, alignment	mean, slope, last
Denoising consistency	residual, update-prediction cosine	mean, slope, last

B.5 Training-Free Health-Score Details

The training-free health score combined standardised trajectory features using fixed weights. For generation i and feature k , the standardised value was

$$z_{ik} = \frac{x_{ik} - \mu_k}{\sigma_k + \epsilon},$$

where x_{ik} is the feature value, and μ_k and σ_k are the dataset mean and standard deviation for that feature. The raw score was

$$\tilde{H}_i = \sum_{k=1}^K w_k z_{ik},$$

where w_k is the fixed feature weight. The final health score was min-max scaled to $[0, 100]$:

$$H_i = 100 \cdot \frac{\tilde{H}_i - \min_j \tilde{H}_j}{\max_j \tilde{H}_j - \min_j \tilde{H}_j + \epsilon}.$$

The weights reflected qualitative expectations about early denoising. The score rewarded moderate latent movement, changing denoiser predictions, agreement between updates and predictions, and some prompt-driven guidance separation. It penalised excessive latent magnitude and high denoising-consistency residuals. These choices made the score interpretable, but they also made it a heuristic rather than an optimised predictor.

Table B.3: Hand-designed weights used in the training-free health score. Positive weights increase the health score; negative weights decrease it.

Feature group	Weight
Latent volatility RMS mean	+1.00
Latent volatility RMS slope	+0.50
Denoiser prediction change RMS mean	+0.75
Update-prediction cosine mean	+0.50
CFG divergence RMS mean	+0.25
Latent RMS mean	−0.25
Latent RMS slope	−0.15
Denoising consistency residual mean	−0.75

B.6 Evaluation Formulae

For prompt-controlled pairwise ranking, all pairs of seeds within a prompt were compared. A pair was concordant when the image with the higher predicted score also had the higher true quality score. Pairwise accuracy was

$$\text{PairwiseAccuracy} = \frac{N_{\text{concordant}}}{N_{\text{concordant}} + N_{\text{discordant}}}.$$

For best-of-five selection, let G_p be the five generations for prompt p , s_i be the predicted score, and q_i be the true quality score. The selected generation was

$$i_p^* = \arg \max_{i \in G_p} s_i.$$

The random baseline was

$$Q_{\text{random}}(p) = \frac{1}{|G_p|} \sum_{i \in G_p} q_i,$$

and the oracle baseline was

$$Q_{\text{oracle}}(p) = \max_{i \in G_p} q_i.$$

The fraction of oracle gain was

$$\frac{\mathbb{E}_p[q_{i_p^*}] - \mathbb{E}_p[Q_{\text{random}}(p)]}{\mathbb{E}_p[Q_{\text{oracle}}(p)] - \mathbb{E}_p[Q_{\text{random}}(p)] + \epsilon}.$$

For abort simulation, a generation was marked as aborted when its early health score was below threshold τ :

$$\text{Abort}_i(\tau) = \begin{cases} 1, & s_i < \tau, \\ 0, & s_i \geq \tau. \end{cases}$$

For the training-free score, $s_i = H_i$. For the learned score, s_i was the target-specific predicted probability.

Since the prefix used five of 25 denoising steps, an aborted generation saved approximately 80% of the sampling steps for that generation.

C Supplementary Results

C.1 Quality Target Diagnostics

The three automatic quality targets used different numerical scales. Table C.1 reports summary statistics for the 36,600 generated images.

Table C.1: Quality score summary statistics for the final generated images. Raw values should be interpreted within each metric rather than compared across metrics.

Target	Mean	SD	Min	5%	25%	Median	75%	Max
CLIP	37.802	4.286	13.187	30.598	34.991	37.949	40.701	53.867
HPSv2	0.309	0.039	0.101	0.241	0.284	0.311	0.337	0.435
ImageReward	1.146	0.644	-2.222	-0.086	0.751	1.297	1.669	2.010

The five seed passes had similar marginal score distributions, suggesting that later prompt-controlled effects were not artefacts of a particular seed pass.

The exact variance-decomposition values behind Figure 4.1 are reported in Table C.2.

Table C.2: Variance decomposition by prompt. Percentages are shown in the main Results; raw variance components are included here for reproducibility.

Target	Between-prompt variance	Within-prompt variance	Between-prompt %	Within-prompt %
CLIP	14.731	3.638	80.2%	19.8%
HPSv2	0.001269	0.000249	83.6%	16.4%
ImageReward	0.325893	0.088499	78.6%	21.4%

C.2 Metric Agreement

The main text reports the key metric-correlation pattern. The heatmaps below provide the full raw and prompt-centred correlation matrices.

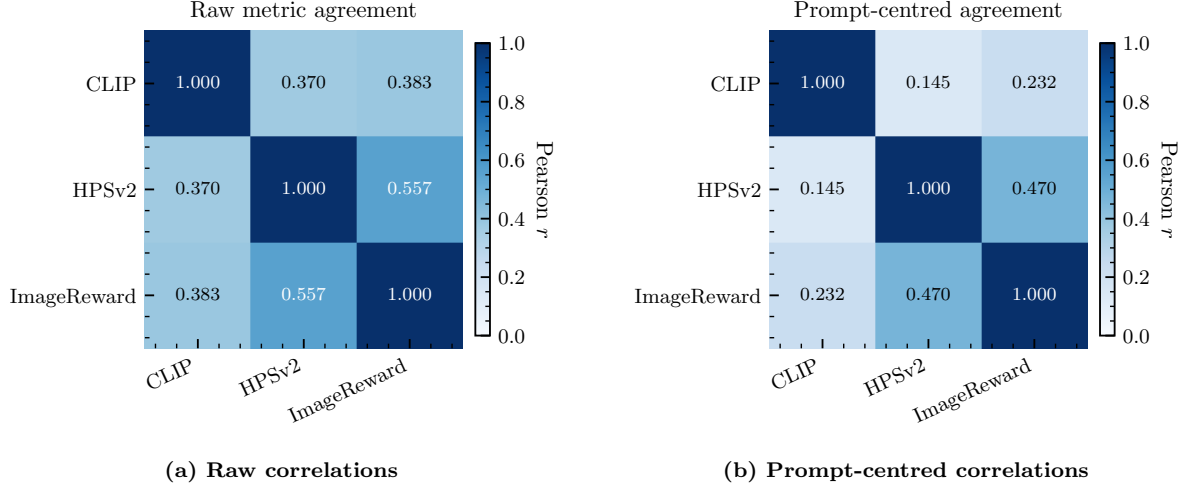


Figure C.1: Metric-agreement heatmaps. Raw correlations include both prompt-level and seed-level variation, while prompt-centred correlations remove each prompt’s mean score.

Table C.3: Within-prompt pairwise agreement between quality metrics.

Metric pair	Usable pairs	Concordant	Discordant	Accuracy
CLIP vs HPSv2	72,868	39,187	33,681	53.8%
CLIP vs ImageReward	73,200	41,195	32,005	56.3%
HPSv2 vs ImageReward	72,868	46,877	25,991	64.3%

C.3 Global Prediction Diagnostics

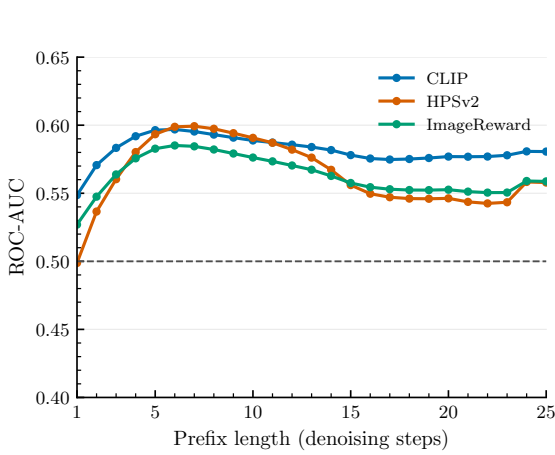
Table C.4 reports the full ROC-AUC and PR-AUC results for global median-split quality prediction.

Table C.4: Global binary quality prediction using the first five denoising steps.

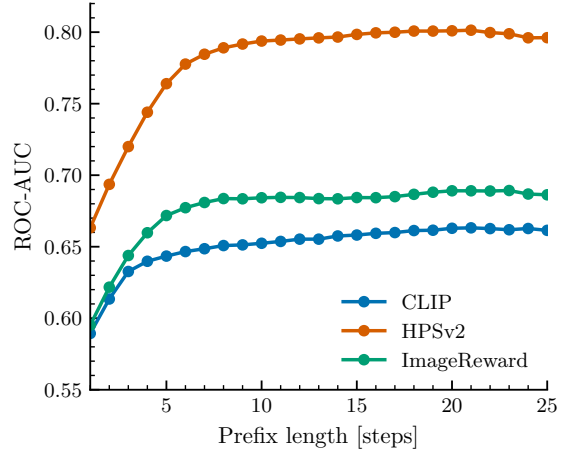
Target	Training-free ROC-AUC	Training-free PR-AUC	Supervised ROC-AUC	Supervised PR-AUC
CLIP	0.596	0.575	0.643	0.622
HPSv2	0.593	0.576	0.764	0.751
ImageReward	0.583	0.562	0.672	0.647

C.4 Prefix-Length Diagnostics

The main text shows the combined prefix-length pattern. The separate plots below provide clearer views of the training-free and supervised results.



(a) Training-free ROC-AUC as a function of prefix length. The hand-designed score peaked around the five-step prefix and declined when later steps were included.



(b) Supervised logistic-regression ROC-AUC as a function of prefix length. The supervised model generally improved as more denoising steps were observed.

Table C.5: ROC-AUC by prefix length. Training-free values use the same hand-designed health-score formula as the prefix-5 global evaluation, recomputed using only the first K denoising steps. Supervised values use logistic regression with compact prefix features at each K .

Target	Score	$K = 1$	$K = 3$	$K = 5$	$K = 10$	$K = 15$	$K = 20$	$K = 25$
CLIP	Training-free	0.549	0.583	0.596	0.589	0.578	0.577	0.581
CLIP	Supervised	0.589	0.633	0.643	0.652	0.658	0.663	0.661
HPSv2	Training-free	0.499	0.560	0.593	0.591	0.556	0.546	0.558
HPSv2	Supervised	0.663	0.720	0.764	0.794	0.798	0.801	0.796
ImageReward	Training-free	0.527	0.564	0.583	0.576	0.558	0.553	0.559
ImageReward	Supervised	0.595	0.644	0.672	0.684	0.684	0.689	0.686

C.5 Prompt-Controlled Diagnostics

Table C.6 reports raw and prompt-centred health-score correlations. Prompt centring removes each prompt’s mean score and therefore tests whether the score tracks seed-level deviations within the same prompt.

Table C.6: Correlations between early health scores and final quality. Prompt-centred correlations remove prompt-level difficulty and test within-prompt seed-level association.

Target	Training-free raw r	Training-free centred r	Learned raw r	Learned Spearman ρ
CLIP	0.198	-0.044	0.314	0.297
HPSv2	0.195	-0.110	0.570	0.551
ImageReward	0.149	-0.045	0.364	0.353

Table C.7 reports grouped cross-validation by prompt. All five generations of a prompt were assigned to the same fold.

Table C.7: Grouped logistic-regression results using prompt identity as the cross-validation group.

Target	Rows	Prompt groups	ROC-AUC	PR-AUC
CLIP	36,600	7,320	0.644	0.623
HPSv2	36,600	7,320	0.764	0.750
ImageReward	36,600	7,320	0.671	0.647

Table C.8: Within-prompt pairwise ranking accuracy.

Target	Score	Usable pairs	Concordant	Pairwise accuracy
CLIP	Training-free health	73,200	35,385	48.3%
CLIP	Learned score	73,200	36,285	49.6%
HPSv2	Training-free health	72,868	33,659	46.2%
HPSv2	Learned score	72,868	43,074	59.1%
ImageReward	Training-free health	73,200	34,974	47.8%
ImageReward	Learned score	73,200	38,826	53.0%

Table C.9: Best-of-five seed selection using training-free and learned scores. Positive oracle-gain fractions indicate improvement over random seed selection.

Target	Selector	Selected quality	Random expected	Oracle best	Fraction of oracle gain
CLIP	Training-free	37.696	37.802	40.104	-4.6%
CLIP	Learned	37.772	37.802	40.104	-1.3%
HPSv2	Training-free	0.3063	0.3090	0.3272	-14.9%
HPSv2	Learned	0.3143	0.3090	0.3272	29.1%
ImageReward	Training-free	1.122	1.146	1.438	-8.1%
ImageReward	Learned	1.177	1.146	1.438	10.8%

C.6 Abort Threshold Diagnostics

The abort simulation used the following definitions. For an early score s_i and threshold τ , the retained set was

$$R(\tau) = \{i : s_i \geq \tau\}.$$

The abort rate was

$$\text{RetainedQuality}_q(\tau) = \frac{1}{|R(\tau)|} \sum_{i \in R(\tau)} q_i,$$

where N is the total number of generations. For each quality target q , retained quality was

$$\text{QualityLift}_q(\tau) = \text{RetainedQuality}_q(\tau) - \frac{1}{N} \sum_{i=1}^N q_i.$$

and quality lift was

$$\text{QualityLift}_q(\tau) = \text{RetainedQuality}_q(\tau) - \frac{1}{N} \sum_{i=1}^N q_i.$$

Since the abort decision was made after five of 25 sampling steps, estimated sampling-step saving was computed as

$$\text{ComputeSaved}(\tau) \approx 0.8 \times \text{AbortRate}(\tau).$$

Table C.10 reports representative training-free abort thresholds. The main text reports a matched-abort-rate comparison between the training-free and learned scores.

Table C.10: Abort simulation using the training-free health score. Compute saved was estimated as 80% of the abort rate because decisions were made after five of 25 denoising steps.

Target	Threshold	Abort rate	Compute saved	Kept quality	Quality lift
CLIP	20	17.5%	14.0%	38.144	+0.342
CLIP	25	31.0%	24.8%	38.296	+0.495
CLIP	30	45.9%	36.7%	38.438	+0.636
CLIP	40	71.7%	57.4%	38.733	+0.931
HPSv2	20	17.5%	14.0%	0.3117	+0.0027
HPSv2	25	31.0%	24.8%	0.3130	+0.0040
HPSv2	30	45.9%	36.7%	0.3145	+0.0055
HPSv2	40	71.7%	57.4%	0.3177	+0.0087
ImageReward	20	17.5%	14.0%	1.184	+0.038
ImageReward	25	31.0%	24.8%	1.201	+0.055
ImageReward	30	45.9%	36.7%	1.220	+0.074
ImageReward	40	71.7%	57.4%	1.252	+0.106

C.7 Feature Diagnostics

Feature diagnostics supported the interpretation that global quality and seed-level quality relied on different trajectory signals. Global quality was most associated with guidance-related features, while HPSv2 seed-level prediction was more associated with early latent stability. Table C.11 repeats the representative patterns from the main Results and situates them as diagnostic, not causal, evidence.

Figure C.3 shows the largest HPSv2 grouped logistic-regression coefficients and provides additional context for the feature patterns in Table C.11.

Table C.11: Representative feature diagnostics. These values summarise the main associations used to interpret the learned model; they should be read as diagnostic associations rather than causal effects.

Setting	Main feature pattern	Association
Global HPSv2	CFG relative divergence	$\rho \approx 0.523$
Prompt-centred HPSv2	Early latent volatility	$\rho \approx -0.30$ to -0.33
Prompt-centred ImageReward	Early latent volatility	$\rho \approx -0.13$
Prompt-centred CLIP	Strongest feature associations	$ \rho \approx 0.05$

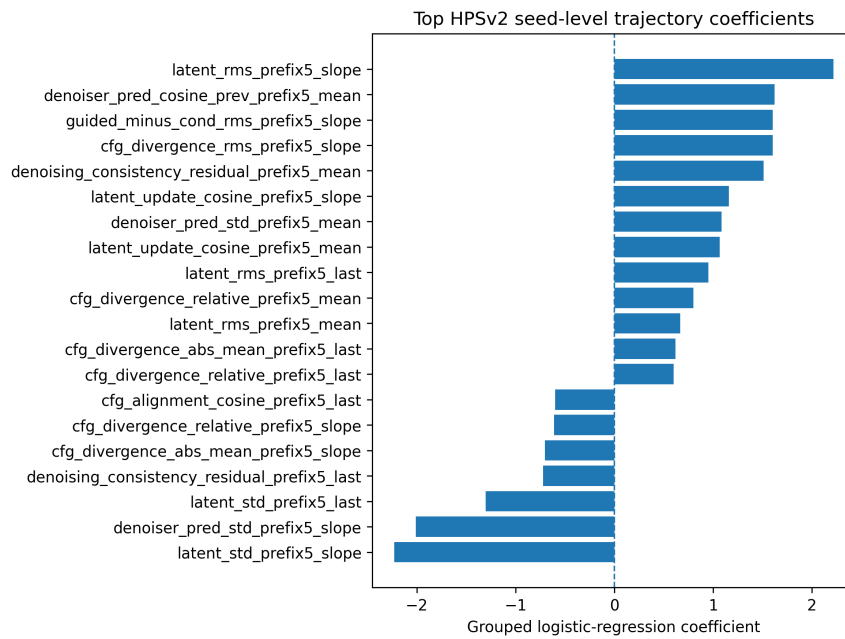


Figure C.3: Largest HPSv2 grouped logistic-regression coefficients by absolute value. The plot provides diagnostic context for which prefix trajectory features contributed most strongly to HPSv2 prediction.

D Additional Prompt-Uncertainty Analysis

As an additional comparison, I analysed prompt-level uncertainty predictions from an external model developed by Yang (2026) in an unpublished manuscript. The model classifies prompts as certain or uncertain before image generation, using only their representations in CLIP text-embedding space. This analysis was not part of the main experimental pipeline, but was included as a supplementary check to compare prompt-level uncertainty with the trajectory-level signals studied in this thesis.

The external dataset contained one prediction for each of the 7,320 prompts used in the main experiment. Each row included a binary prediction, **certain** or **uncertain**, together with the probabilities **prob_certain** and **prob_uncertain**. These prompt-level predictions were merged with the 36,600 generated images from the main experiment using **prompt_id**. The merge therefore associated each generated image with the uncertainty score of its original prompt.

Table D.1: Summary of the prompt-uncertainty merge.

Quantity	Value
External prompt-level rows	7,320
Trajectory/generation rows	36,600
Merged generation rows	36,600
Unique prompts after merge	7,320
Prompts labelled uncertain	5,832
Prompts labelled certain	1,488
Percentage labelled uncertain	79.7%

The first analysis examined whether the external prompt-uncertainty score was associated with final image quality at the generation level. Correlations were computed between **prob_uncertain** and the three final quality scores used in the thesis: CLIP, HPSv2, and ImageReward.

Table D.2: Generation-level correlations between prompt uncertainty and final image quality.

Quality target	Pearson r	Spearman ρ
CLIP	-0.033	-0.045
HPSv2	-0.042	-0.048
ImageReward	0.003	0.015

All correlations were close to zero. This indicates that prompts assigned higher uncertainty by the external CLIP-text model did not consistently produce lower-quality images according to the automatic evaluation metrics used in the main thesis.

Because the main results showed that most quality variation occurred at the prompt level, I also repeated the analysis after averaging image quality across the five seeds for each prompt. This tested whether prompt uncertainty was related to mean prompt difficulty.

The prompt-level correlations were also weak. The strongest relationship was a small negative correlation with HPSv2, but its magnitude was too low to suggest that the external prompt-uncertainty score explains prompt difficulty in this dataset.

I additionally tested whether prompt uncertainty could identify below-median quality generations. For each quality metric, generations below or equal to the median were treated as the positive class, and **prob_uncertain** was used as the prediction score.

The ROC-AUC values were close to chance. This further supports the interpretation that the external prompt-uncertainty score does not act as a strong proxy for final image quality in the present experiment.

Table D.3: Prompt-level correlations between prompt uncertainty and mean final image quality.

Mean quality target	Pearson r	Spearman ρ
CLIP	-0.037	-0.050
HPSv2	-0.046	-0.053
ImageReward	0.003	0.017

Table D.4: ROC-AUC for using prompt uncertainty to identify below-median quality generations.

Prediction task	ROC-AUC
Below-median CLIP	0.524
Below-median HPSv2	0.521
Below-median ImageReward	0.494

Finally, I examined whether prompt uncertainty was related to within-prompt seed variability. This matters because the thesis distinguishes global prompt difficulty from the harder task of ranking seeds within the same prompt.

Table D.5: Correlations between prompt uncertainty and within-prompt quality variation.

Within-prompt standard deviation	Pearson r	Spearman ρ
CLIP	-0.023	0.001
HPSv2	0.003	0.016
ImageReward	-0.021	-0.025

Prompt uncertainty showed almost no relationship with within-prompt quality variation. This suggests that prompts considered uncertain in CLIP text-embedding space were not systematically more variable across random seeds.

Overall, this supplementary analysis suggests that prompt-level uncertainty and trajectory-level quality signals capture different phenomena. The external model developed by Yitong et al. identifies uncertainty before generation begins, based on the geometry of CLIP text embeddings. In contrast, the main analysis in this thesis examines uncertainty-like behaviour during the denoising process itself. In this dataset, the two forms of uncertainty were only weakly related, suggesting that prompt-level uncertainty may be complementary to trajectory-level analysis rather than a replacement for it.